

THE TRIADIC AGENT IN CONTEXT: TOWARD A THEORY OF DISTRIBUTED MORAL AGENCY AND SOCIAL COGNITION

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ABSTRACT

The Triadic Theory of Action (Selection, Identification, Anticipation) and the JARC framework have thus far been developed primarily at the level of the individual agent. This paper extends these foundations to the **social and systemic level**. We argue that moral judgment, responsibility, and anticipation are not properties of isolated agents but emerge from the **structural coupling** of multiple agents within networks of practice. We introduce the concept of **Distributed Triadic Action**—a model in which Selection, Identification, and Anticipation are distributed across agents, institutions, and artifacts. We propose that **social objectivity** is the invariance of distributed acts under the expansion of the social network, and that **systemic responsibility** arises from the architecture of the network rather than from individual intentionality. The paper addresses the challenges of AI governance, multi-agent systems, and the alignment of human and machine values, offering a framework that is neither individualist nor collectivist, but **relational and processual**.

Keywords: Distributed Triadic Action, Systemic Responsibility, Social Objectivity, Multi-Agent Systems, AI Governance, Networked Agency, Structural Coupling.

1. INTRODUCTION: THE LIMITS OF THE INDIVIDUALIST FRAMEWORK

Our prior work (Papers I–IX) has developed a comprehensive theory of cognitive and moral agency grounded in the triadic act of the individual agent. This framework has proven powerful in explaining judgment, error, responsibility, and ethics as properties of a single acting system.

However, the real world is not composed of isolated agents. Human cognition, moral judgment, and practical action are always embedded in **social, institutional, and technological contexts**. An individual's anticipation depends on the anticipations of others; the identification of a situation is shaped by shared categories; the selection of a course of action is constrained by social norms and legal frameworks.

To ignore this social dimension is to risk reducing the triadic framework to a form of methodological individualism—a limitation that has plagued both philosophy and cognitive science. This paper addresses this gap by extending the triadic theory to the **level of networks, institutions, and distributed cognition**.

We begin by examining the problem of distributed action: when multiple agents act together, who is responsible for the outcome? We then show how the triadic phases can be distributed across a network, and we introduce the concepts of social objectivity and systemic responsibility. Finally, we discuss the implications for AI governance and multi-agent systems, and we propose a new type of responsibility—**Mesh Responsibility**—that captures the networked nature of contemporary agency.

2. THE SOCIAL NATURE OF THE TRIADIC ACT

2.1 From Individual to Distributed Triad

We propose that the triadic act (Selection, Identification, Anticipation) is not only performed by individual agents but can also be **distributed across multiple agents and artifacts**.

Triadic Phase	Individual Level	Distributed Level
Selection	An agent focuses attention on a relevant fragment.	A team, organization, or algorithm collectively filters information.
Identification	An agent categorizes the selected fragment.	A shared ontology, database, or legal code defines the category.
Anticipation	An agent simulates possible futures and commits.	A network of agents simulates scenarios collectively; institutions encode expectations.

Definition 1: Distributed Triadic Action (DTA) is a process in which the three phases of the triadic act are performed by different agents or systems, yet function as a **single, coherent cognitive unit** through structural coupling and communication.

2.2 Structural Coupling

Following Maturana and Varela, we define **structural coupling** as the ongoing, reciprocal interaction between systems that leads to mutual adaptation. In the context of DTA, structural coupling ensures that:

- The Selection of one agent influences the Identification of another.
- The Anticipation of one agent constrains the Anticipation of others.
- The commitments of individual agents become **institutional facts** (Searle) that shape the environment for all.

This coupling is not a side effect; it is the **constitutive condition** of social cognition. Without it, there is no society, no language, no law—only isolated monads.

3. SOCIAL OBJECTIVITY AS INVARIANCE OF THE DISTRIBUTED ACT

3.1 The Problem of Social Objectivity

In Paper II, we defined objectivity as the invariance of an act under the expansion of practice. At the individual level, this meant that a concept or judgment was objective if it retained its success across diverse contexts.

At the social level, we extend this definition:

Definition 2: Social Objectivity is the invariance of a Distributed Triadic Action under the expansion of the social network—that is, when new agents, institutions, or practices are included, the distributed act remains effective.

3.2 Examples

- **Scientific objectivity:** A scientific theory is socially objective if it retains its predictive and explanatory power when new researchers, instruments, or experimental paradigms are introduced.
- **Legal objectivity:** A law is socially objective if it maintains its regulatory effectiveness when applied to new cases, in new jurisdictions, or under new social conditions.
- **Moral objectivity:** A moral norm is socially objective if it preserves the ontological integrity of agents across diverse social contexts.

This criterion provides a **non-relativist foundation for social norms** without appealing to metaphysical absolutes.

4. SYSTEMIC RESPONSIBILITY

4.1 The Limitations of Individual Responsibility

Functional Responsibility (Paper VII) was defined as the capacity of an individual system to recognize the causal link between its selection and the outcome of its anticipation. However, in distributed systems—such as corporations, AI networks, or hybrid human-machine teams—responsibility cannot be reduced to individual acts.

Definition 3: Systemic Responsibility is the capacity of a distributed system to recognize and regulate the causal links between its collective selections, identifications, and anticipations, and the outcomes they produce.

4.2 Mechanisms of Systemic Responsibility

Mechanism	Description
Traceability	The network must be able to trace which agent or subsystem contributed to which phase of the distributed act.
Accountability protocols	The network must have procedures for identifying and correcting failures in anticipation or commitment.
Feedback loops	The network must integrate outcomes into future distributed acts, adjusting its collective anticipation.

4.3 The Electronic Liability Fund (ELF) Revisited

In Paper VIII, we proposed the Electronic Liability Fund as a mechanism for financial accountability of autonomous systems. At the social level, the ELF can be extended to **distributed liability**: when a network of agents produces a harmful outcome, the ELF is debited proportionally to the contribution of each agent or subsystem to the distributed act.

5. THE TRIADIC AGENT IN AI GOVERNANCE

5.1 The Problem of Multi-Agent AI Systems

As AI systems become more interconnected—swarms of drones, autonomous vehicles communicating with infrastructure, large language models coordinating with other models—the traditional model of individual accountability breaks down. Who is responsible when a swarm of drones makes a collective decision?

Our framework offers a solution: **responsibility is not located in any single agent but in the architecture of the distributed network.**

5.2 Governance Implications

Challenge	Triadic Solution
Cascading failures	Traceability mechanisms identify which Selection, Identification, or Anticipation failed.
Value alignment	Social objectivity ensures that the distributed act remains effective across diverse human values.
Regulatory oversight	Systemic responsibility can be audited through protocols that assess the network's capacity for traceability and feedback.

6. TOWARD A THEORY OF DISTRIBUTED MORAL AGENCY

6.1 Beyond Individualism and Collectivism

The extension of the triadic framework to the social level has profound implications for philosophy, cognitive science, and AI governance:

- Beyond individualism:** Cognition and morality are not properties of isolated minds but of **relational systems**.
- Beyond collectivism:** Society is not a monolithic entity but a **network of distributed triadic acts**.
- Beyond relativism:** Social objectivity provides a verifiable criterion for evaluating social norms and institutions.

This is not a rejection of our earlier work but its **maturation**. The triadic act is not confined to the skull; it is **embodied, embedded, and extended** (Clark & Chalmers) into the social and technological world.

6.2 Mesh Responsibility: A New Concept

We introduce the term **Mesh Responsibility** to capture the distributed, networked nature of accountability in contemporary socio-technical systems. Mesh Responsibility is the property of a network in which each node (agent, institution, or artifact) contributes to the triadic act, and the network as a whole is held accountable for the outcomes of its distributed cognition.

Mesh Responsibility differs from individual responsibility in that it:

- Is **non-localizable**: it cannot be pinned on a single agent.
- Is **proportional**: each node's share of responsibility is proportional to its causal contribution.
- Is **dynamic**: responsibility shifts as the network reconfigures.

This concept provides a practical framework for governance of AI ecosystems, supply chains, and hybrid human-machine collectives.

7. RELATION TO EXISTING THEORIES OF DISTRIBUTED COGNITION

Our framework engages critically with the tradition of distributed cognition (Hutchins, 1995; Clark & Chalmers, 1998). We agree that cognition is not confined to the individual brain but is distributed across people, tools, and environments. However, we diverge in two important respects:

1. **Teleological grounding**: While Hutchins focuses on the functional organization of cognitive systems, we emphasize that distributed cognition is always **goal-directed** and **anticipatory**. The triadic act provides the teleological engine that drives distributed systems.
2. **Responsibility attribution**: Clark and Chalmers argue for the "extended mind," but they do not address the question of responsibility. Our framework fills this gap by providing criteria for attributing systemic responsibility and mesh responsibility.

We also build on Maturana and Varela's concept of structural coupling, but we extend it by specifying the triadic phases as the content of the coupling. This makes the concept operational for engineering and governance.

8. CONCLUSION: THE NETWORKED ACT

This paper has extended the Triadic Theory of Action and the JARC framework to the social and systemic level. We have introduced Distributed Triadic Action, Social Objectivity, Systemic Responsibility, and Mesh Responsibility as concepts that bridge the individual and the collective.

The future of AI governance, legal theory, and social epistemology will depend on our ability to understand not only how individual agents act but how **networks of agents act together**. The triadic framework, now extended to the social dimension, provides the conceptual tools for this task.

The act is not alone. It is always part of a **network of acts**. And the network, too, must be judged.

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